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Stacked package structure and stacked packaging method for chip

Abstract

A stacked package structure for a chip, can include: a substrate having a first surface and a second surface opposite thereto; a first die having an active and back faces, where the active face of the first die includes pads; a first enclosure that covers the first die; an interlinkage that extends to the first enclosure to electrically couple with the pads; a first redistribution body electrically coupled to the interlinkage, and being partially exposed on a surface of the stacked package structure to provide outer pins for electrically coupling to external circuitry; a penetrating body that penetrates the first enclosure and substrate; a second die having an electrode electrically coupled to a first terminal of the penetrating body; and a second terminal of the penetrating body that is exposed on the surface of the stacked package structure to provide outer pins for electrically coupling to the external circuitry.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of the following application, U.S. patent application Ser. No. 16/913,096, filed on Jun. 26, 2020, which is a continuation of U.S. patent application Ser. No. 15/283,573, filed on Oct. 3, 2016, now issued as U.S. Pat. No. 10,763,241, and which are both hereby incorporated by reference as if they are set forth in full in this specification, and which also claims the benefit of Chinese Patent Application No. 201510665364.6, filed on Oct. 15, 2015, which is also incorporated herein by reference in its entirety.

FIELD OF THE INVENTION
(1) The present disclosure generally relates to the field of chip packaging, and more particularly to stacked chip package structures and associated manufacturing methods.
BACKGROUND
(2) Integrated circuit dice are typically packaged prior to being integrated with other electrical elements or devices in the manufacturing process. The package structure may at least provide sealing of dice, as well as provide electrical connectivity ports to external circuitry. For example, the package structure can provide electrical connectivity between dice and base board of electrical

or electronic products, protection from pollution, mechanical support, heat dissipation, and also may reduce heat mechanical strain.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a cross-sectional view diagram of a first example stacked package structure, in accordance with embodiments of the present invention.

(2) FIG. **2** is a cross-sectional view diagram of a second example stacked package structure, in accordance with embodiments of the present invention.

(3) FIG. **3** is a cross-sectional view diagram of a third example stacked package structure, in accordance with embodiments of the present invention.

(4) FIG. **4** is a flow diagram of an example method of making a stacked package structure, in accordance with embodiments of the present invention.

DETAILED DESCRIPTION

(5) Reference may now be made in detail to particular embodiments of the invention, examples of which are illustrated in the accompanying drawings. While the invention may be described in conjunction with the preferred embodiments, it may be understood that they are not intended to limit the invention to these embodiments. On the contrary, the invention is intended to cover alternatives, modifications and equivalents that may be included within the spirit and scope of the invention as defined by the appended claims. Furthermore, in the following detailed description of the present invention, numerous specific details are set forth in order to provide a thorough understanding of the present invention. However, it may be readily apparent to one skilled in the art that the present invention may be practiced without these specific details. In other instances, well-known methods, procedures, processes, components, structures, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the present invention.

(6) Connections between dice or integrated circuits and pins of a package structure is an essential portion of achieving input and output connections between dice and external circuitry. Stacked chip packaging techniques are becoming more widely used, in order to reduce the size of an integrated circuit. In some cases, bonding wires can be used to achieve such connectivity. High purity thin metal wire (e.g., gold wire, copper wire, aluminum wire, etc.) can be employed to connect pads of dice to a leadframe or printed-circuit board (PCB). However, there may be drawbacks associated with the employment of bonding wires, such as pad cratering, tail inconformity, bending fatigue, vibration fatigue, breakage, and disconnection.

(7) Stacked packaging techniques continue to develop so as to meet the requirements and challenges of various semiconductor processes, and new materials based on the relationship of manufacturing and packaging. Thus, more stable and reliable packaging methods may be developed in order to connect internal chips with external pins. However, heat dissipation, package size, and package height may be compromised in some cases.

(8) In one embodiment, a stacked package structure for a chip, can include: (i) a substrate having a first surface and a second surface opposite thereto; (ii) a first die having an active face and a back face opposite thereto, where the first die is arranged above the first surface of the substrate, the back face of the first die is relatively close to the first surface of the substrate, and the active face of the first die includes pads; (iii) a first enclosure that covers the first die; (iv) at least one interlinkage that extends to the first enclosure to electrically couple with the pads; (v) at least one first redistribution body electrically coupled to the interlinkage, and being partially exposed on a surface of the stacked package structure to provide outer pins for electrically coupling to external circuitry; (vi) at least one penetrating body that penetrates the first enclosure and the substrate; (vii) a second die having at least one electrode electrically coupled to a first terminal of the penetrating

body; and (viii) a second terminal of the penetrating body that is at least partially exposed on the surface of the stacked package structure to provide outer pins for electrically coupling to the external circuitry.

(9) In particular embodiments, a stacked package structure can utilize an interlinkage, a redistribution body, and a penetrating body, without bonding wires, in order to achieve electrical conductivity between pads and a leadframe or PCB board. Referring now to FIG. **1**, shown is a cross-sectional view diagram of a first example stacked package structure, in accordance with embodiments of the present invention. Package structure **10** can include substrate **110**, die **210**, enclosure **311**, spaced interlinkages **310**, redistribution bodies **410**, spaced penetrating bodies **510**, and die **610**. Substrate **110** may include semiconducting material (e.g., silicon, germanium, indium antimonide, gallium arsenide, indium arsenide, gallium nitride, etc.), insulating material (e.g., epoxy resin, polyester glass fiber, silicon dioxide, polytetrafluorethylene, glass, ceramic, etc.), or various combinations thereof.

(10) Package substrate **110** may include "first" and "second" surfaces. Dice **210** and **610** may each include an active face and a back face opposite thereto. Device layers of die **210** and die **610** may lie on the active face, and the device layer may include a transistor, and other devices (e.g., resistors, capacitors, inductors, etc.). A plurality of metal layers may lie above the device layer. Each of the metal layers may include metal interlinkages made from copper, and through holes for electrical connections between the metal interlinkages. The metal interlinkages and through holes can be surrounded by insulated interlayer dielectric. A plurality of pads may be arranged above the metal layers. For example, pad **211** can be located on the active face of die **210**, while pads on die **610** are not shown in the diagram.

(11) Die **210** may be arranged above the first surface of substrate **110**. For example, the back face of die **210** can be relatively close to the first surface of substrate **110**, and may be pasted to the first surface of substrate **110** via adhesive layer **111**. Adhesive layer **111** may be insulating glue, or Au—Si alloy, Pb—Sn alloy, Sn—Ag—Cu alloy, and/or conductive glue (e.g., conductive glue with conductive particles, dispersant added into the base of the epoxy resin, etc.). Enclosure **311** may be formed above the first surface of die **210** and substrate **110** for covering and protecting die **210** from being damaged and polluted. Enclosure **311** may be formed by any suitable type of material (e.g., ceramic, epoxy resin, etc.). Enclosure **311** can be provided with a plurality of "first" openings corresponding to pads **211**, in order to expose pad **211**. For example, laser etching or mechanical drilling may be employed to form such first openings.

(12) Interlinkages **310** may be used to lead out electrodes of die **210**. Each of interlinkages **310** can be extended into enclosure **311**, so as to electrical coupled to pad **211** of die **210**. In this example, each of the interlinkages **310** may include a "first" portion extending on the surface of **311**, and a "second" portion of enclosure **311** extending to a corresponding pad **211**. Redistribution body **410** may be used to redistribute positions of electrodes on the interlinkages **310**. Each of redistribution body **410** can be electrically coupled to one interlinkage **310** and partially exposed on the surface of stacked package structure **10**, and may be used as outer pins for providing external electrical connectivity.

(13) In this particular example, redistribution body **410** may include a first portion extending on the surface of enclosure **311**, a second portion extending from the surface of enclosure **311** to the second surface of substrate **110**, and a "third" portion extending on the second surface of substrate **311**. The first portion of redistribution body **410** can be extended to the first portion of interlinkage **310**, in order to achieve electrical connection with interlinkage **310**. The third portion of redistribution body **410** may be partially exposed on the surface of stacked package structure **10**, to serve as outer pins for external electrical connectivity. Thus, die **210** can be electrically coupled external circuitry.

(14) Each of the penetrating bodies **510** may penetrate enclosure **311** and substrate **110**. The first terminal of penetrating body **510** may be electrically coupled to die **610**, and the second terminal

can be partially exposed on the surface of the stacked package structure as outer pins for electrically coupling to external circuitry. Thus, die **610** may be electrically coupled to such external circuitry. In this particular example, the first terminal of penetrating body **510** can extend on the surface of enclosure **311**, the second terminal may extend on the second surface of substance **110**, and the mid-section can extend from the surface of enclosure **311** to the second surface of substrate **110**.

(15) The active face of die **610** can face enclosure **311**, and at least one electrode on die **610** may be electrically coupled to the first terminal of penetrating body **510** via conductive bump **611**. In this way, the electrodes on die **610** may be lead out to the surface of stacked package structure **610** via penetrating body **510**, for electrically coupling to external circuitry. At least one electrode on die **610** can be electrically coupled to the first portion of interlinkage **310** via conductive bump **611**, so as to electrically couple the electrodes of die **210** and die **610** inside stacked package structure **10**. In this way, the number of outer pins of the stacked package structure can be reduced.

(16) In other examples, die **610** may be packaged by wire bonding, such as the back face of die **610** may be relatively close to and above the first portion of interlinkage **310**. For example, the back face of die **610** may be pasted on interlinkage **310** via insulation glue. At least one electrode on die **610** can be electrically coupled to the first terminal of penetrating body **510** via conductive wires. In this way, the electrodes on die **610** may lead to the surface of the stacked package structure through the second terminal of penetrating body **510**, in order to provide outer pins for electrically coupling to external circuitry. Moreover, at least one electrode on die **610** may be coupled to the first portion of interlinkage **310** through conductive wires, so as to electrically couple die **110** with die **610** inside stacked package structure **10**.

(17) The first and second portions of interlinkage **310**, the first to third portions of redistribution body **410**, the first and second terminals of penetrating body **510** as well as the mid-section thereof, may be made from the same or different conducting materials. In some cases, one or more such portions may be selectively made from the same materials at substantially the same time. For example, the first and second portions of interlinkage **310**, the first and second portions of redistribution body **410**, the first terminal and the mid-section of penetrating body **510** can be formed by a patterned conducting layer. For example, conducting layer **41** may include a first portion extending on the surface of the first enclosure, a second portion extending from the surface of enclosure **311** to the inside and electrically coupling to pad **211**, and a third portion extending from the surface of enclosure **311** to the second surface of substrate **110**. Conducting layer **41** may include metal layer **411**, and metal layer **412** located above metal layer **411**. Metal layers **411** and **412** may be formed from any appropriate metal materials (e.g., Ni, Al, Ti, W, Pt, Cu, Au, Co, Ta, etc.) or alloy materials (e.g., TiN, TiW, etc.).

(18) For example, conducting layer **41** may be formed by forming a plurality of first openings that extend from the surface of enclosure **311** to pads **211**, and a plurality of second openings that extend from the surface of enclosure **311** to substrate **110** by laser etching or mechanical drilling. In some cases, conducting layer **51** may be patterned on the second surface of substrate **110** prior to forming the openings, so as to form the third portion of redistribution body **410** and the second terminal of penetrating body **510**. Conducting layer **51** can be partially exposed on the surface of stacked package structure **10**, in order to provide outer pins for electrically coupling to external circuitry. Conducting layer **51** may include metal layer **511** formed on the second surface of substrate **110**, and solder layer **512** located on metal layer **511**. Solder layer **512** may be formed from solder materials, such as W. Solder layer **512** may be used to electrically couple to the lead frame, PCB, or other electronic devices. Metal layers **411**, **412**, and **511** may be made from any appropriate metal materials (e.g., Ni, Al, Ti, W, Pt, Cu, Au, Co, Ta, etc.) or alloy materials (e.g., TiN, TiW, etc.).

(19) Stacked package structure **10** may also include enclosure **312** that can protect the stacked package structure from being damaged and/or polluted by covering die **610**. Enclosures **311** and

312 may be made from different types of materials (e.g., ceramic, epoxy resin, etc.). In certain embodiments, the stacked package structure may be adapted to a chip package with a relatively high space density between pads, by firstly leading out the electrodes via the interlinkages and the first redistribution bodies when packaging the first die. In such a case, the bonding wires may not be needed and the package resistors can be reduced. Secondly, leading out the electrodes on the second die via the penetrating bodies that penetrates the first enclosure and the substrate in a similar way after packaging the first die, so as to package the chip in a stack with reduced packaging size and fewer pins.

(20) Referring now to FIG. **2**, shown is a cross-sectional view diagram of a second example stacked package structure, in accordance with embodiments of the present invention. Stacked package structure **20** can include substrate **120**, die **220**, enclosure **321**, spaced interlinkages **320**, enclosure **322**, spaced redistribution bodies **420**, spaced penetrating bodies **520**, die **620**, patterned conducting layer **521**, and at least one redistribution body **720**. In this particular example, stacked package structure **20** can also include conducting layer **521** and redistribution body **720**. Patterned conducting layer **521** can be formed on the first surface of the substrate **120**. The back face of die **220** may be electrically coupled to conducting layer **521** via conductive stick glue **121**, such that the electrodes on the back face of die **220** are lead to conducting layer **521**.

(21) In some cases, interlinkage **320** may include a first portion that extends on the surface of enclosure **321**, and a second portion that extends to the inside of enclosure **321** and electrically couples to pad **221**. Redistribution body **420** may include a first portion that extends on the surface of enclosure **321**, a second portion that extends from the surface of enclosure **321** to the second surface of substrate **120**, and a third portion that extends on the second surface of substrate **120**. The third portion can be partially exposed on the surface of stacked package structure **20** in order to provide outer pins for electrically coupling the electrodes on the active face of die **220** to external circuitry. Penetrating body **520** may have a first terminal that extends on the surface of enclosure **321**, a second terminal that extends on the second surface of substrate **120**, and a mid-section that extends from the surface of enclosure **321** to the second surface of substrate **120**.

(22) Die **620** may be fabricated in stacked package structure **20** in substantially the same way. For example, die **620** can be flip-chip mounted (e.g., electrically coupled by conductive bumps) or be front-chip mounted (e.g., electrically coupled by conductive wires). Also, electrodes on die **620** can be electrically coupled to the first terminal of penetrating body **520** via conductive bumps **621** or conductive wires. The second terminal of penetrating body **520** can be partially exposed on the surface of stacked package structure **20**, in order to provide outer pins for electrically coupling the electrodes on the active face of die **620** to external circuitry. At least one electrode of die **620** may be electrically coupled to the first portion of the interlinkage **320** via conductive bumps **621** or conductive wires. In the example of FIG. **2**, chip **620** is flip-chip mounted, thus the electrodes of die **220** and die **620** are electrically coupled inside stacked package structure **20**.

(23) Redistribution body **720** may include a first portion that extends on the second surface of substrate **120**, and a second portion that extends from the second surface of substrate **120** to the surface of conducting layer **521**, and may be electrically coupled to the electrodes on the back face of die **22**. The first portion of redistribution body **720** can be partially exposed on the surface of stacked package structure **20**, in order to provide outer pins for electrically coupling the electrodes on the back face of die **220** to external circuitry. In this example, the second portion of redistribution body **420** can include a first conducting channel that extends from the surface of enclosure **321** to the surface of conducting layer **521**, and a second conducting channel that extends from the surface of conducting layer **521** to the second surface of substrate **120**. The mid-section of penetrating body **520** may include a third conducting channel that extends from the surface of enclosure **321** to conducting layer **521**, and another conducting layer that extends from the surface of conducting layer **521** to the second surface of substrate **120**.

(24) In this example, the first and second portions of interlinkages **320**, the first portion and the first

conducting channel of redistribution body **420**, the first terminal and the third conducting channel of penetrating body **520** may be formed by patterned conducting layer **42**. Conducting layer **42** may include a first portion that extends on the surface of enclosure **321**, a second portion that extends from the surface of enclosure **321** to pad **221**, and a third portion that extends from the surface of enclosure **321** to the surface of conducting layer **521**.

(25) For example, a method of forming conducting layer **42** can include forming a plurality of first openings that extends from the surface of enclosure **321** to pads **221** and a plurality of second openings that extends from the surface of enclosure **321** to the surface of conducting layer **521** by laser etching or mechanical drilling the surface of enclosure **321**. The method can also include forming the conducting material layer by plating or depositing on the surface of enclosure **321**, the first openings and the second openings, and etching the conducting material layer by a mask, so as to obtain patterned conducting layer **42**. The second conducting channel, the fourth conducting channel, and the second portion of redistribution body **720** may be formed before conducting layer **42**.

(26) The second conducting channel, the fourth conducting channel, and the second portion of redistribution body **720** can be made from the same conducting material **522** at substantially the same time. This can include forming patterned conducting layer **521** on the first surface of substrate **120**, and forming patterned conducting layer **52** as the outer pins of stacked package structure **20** before die **220** can be mounted to substrate **120**. Thus, conducting layer **52** may include the third portion of redistribution body **420**, the second terminal of penetrating body **520**, and the first portion of the second redistribution body. The method can also include forming a plurality of openings that extends from the surface of conducting layer **521** to the second surface of substrate **120** by conducting the opening process on the surface of conducting layer **521**. The method can also include forming the second conducting channel, the fourth conducting channel, and the second portion of redistribution body **720**, which are electrically coupled to parts of patterned conducting layer **52** by filling or depositing the conducting material in the openings.

(27) In this particular example, conducting layer **42** may include metal layer **421** that extends on the surface of enclosure **321** for thickening conducting layer **42**. Conducting layer **42** may also include metal layer **422** disposed on the surface of metal layer **421**, and inside the first and second openings, as well as metal layer **423** disposed on metal layer **422**. Metal layer **422** may be formed by a metal seed layer of the metal layer **423** formed by plating. Conducting layer **52** may include metal layer **523** disposed on the second surface of substrate **120**, and solder layer **524** disposed on metal layer **523**. This particular example may be suitable for packaging the first die with electrodes on the back face. In such a case, the electrodes on the back face of the first die can be lead out via the second redistribution body, in order to provide outer pins for electrically coupling to external circuitry.

(28) Referring now to FIG. **3**, shown is a cross-sectional view diagram of a third example stacked package structure, in accordance with embodiments of the present invention. Stacked package structure **30** may include package substrate **130**, die **230**, adhesive layer **131**, spaced interlinkages **330**, enclosure **331**, spaced redistribution bodies **430**, spaced penetrating bodies **530**, enclosure **322**, die **630**, and enclosure **333**. The back face of die **230** may be mounted to a first surface of substrate **130** that also has a second surface opposite thereto. Enclosure **331** may cover die **230**. Interlinkage **330** may include a first portion that extends on the surface of enclosure **331**, and a second portion that extends from the surface of enclosure **331** to pads **231** on the active face of die **230**. The first portion of interlinkage **330** can be partially exposed on the surface of stacked package structure **30**, in order to provide pins for electrically coupling die **230** to external circuitry. Enclosure **332** may cover interlinkage **330**, in order to protect interlinkage **330** from possibly being damaged or polluted.

(29) Redistribution body **430** may include a first portion that extends on the surface of enclosure **332**, and a second portion that extends from the surface of enclosure **332** to the inside and may be

electrically coupled with the first portion of interlinkage **430**. Penetrating body **530** may have a first terminal that extends on the second surface of substrate **130**, a second terminal including a first portion that extends on the surface of enclosure **332** and a second portion that extends to the inside of enclosure **332**, and a mid-section that extends from the second terminal of penetrating body **530** to the second surface of substrate **130** and is electrically coupled to the first terminal of penetrating body **530**. In this particular example, the mid-section of penetrating body **530** may further include a first portion that extends on the surface of the first enclosure and a second portion that extends on the second surface of substrate **130**.

(30) Interlinkage **330** and the mid-section of the penetrating body can be formed by patterned conducting layer **43**. Conducting layer **43** may include a first portion that extends on the surface of enclosure **331**, a second portion that extends from the surface of enclosure **331** to pad **231**, and a third portion that extends from the surface of enclosure **331** to the second surface of substrate **130**. The process of forming patterned conducting layer **43** may include forming a plurality of first openings that extends from the surface of enclosure **331** to the pads **231**, and a plurality of second openings that extends from the surface of enclosure **331** to the second surface of substrate **130** by conducting opening process on the surface of enclosure **331** by laser etching or mechanical drilling. The second openings can be terminated at patterned conducting layer **632** on the second surface of substrate **130**. The process can also include forming the conducting material layer by plating or depositing on the surface of enclosure **331**, the first opening and the second opening. The process can also include forming patterned conducting layer **43** by etching the conducting material layer by a mask. Conducting layer **43** may include metal layer **431** formed by plating the seed layer and metal layer **432** on metal layer **431**.

(31) Conducting layer **632** may be formed on the second surface of substrate **130**, in order to serve as the first terminal of penetrating body **530**. The active face of die **630** can be facing toward conducting layer **632**, and electrodes on the active face of die **630** can be electrically coupled to the conducting regions of patterned conducting layer **632** via conductive bumps **631**, so that the electrodes of die **630** are lead to conducting layer **632**, and further to the second terminal of penetrating body **530** through the mid-section of penetrating body **530**, where the conducting regions are isolated from each other. The second terminal of penetrating body **530** may be partially exposed on the surface of stacked package structure **30**, in order to provide outer pins for electrically coupling die **630** to external circuitry.

(32) Die **630** may also be packaged in the way of wire bonding, such as where the back face of die **630** may be pasted to conducting layer **632** by insulating glue. The electrodes on die **630** can be electrically coupled to the first terminal (e.g., conducting layer **632**) of penetrating body **530** via conductive wires such that the electrodes on die **630** are lead to the surface of the stacked package structure through the second terminal of penetrating body **530**, in order to provide outer pins for electrically coupling to external circuitry. In addition, in stacked package structure **30**, the first portion of redistribution body **430** can be electrically coupled to the second terminal of penetrating body **530**, so as to electrically couple die **130** to die **630**, in order to reduce the number of pins of stacked package structure **30**. For example, the first portion of redistribution body **430** may extend to the second terminal of penetrating body **530** to establish electrical connectivity therebetween.

(33) In this particular example, redistribution body **430** and the second terminal of penetrating body **530** can be formed by patterned conducting layer **53**. Conducting layer **53** may include a first portion that extends on the surface of enclosure **332**, a second portion that extends to the inside of enclosure **332** and electrically couples to the mid-section of penetrating body **530** and the first portion of interlinkage **430**. The method of forming conducting layer **53** may include forming a plurality of openings that extends to interlinkage **430** and a plurality of openings that extends to the mid-section of penetrating body **530** by conducting opening process on the surface of enclosure **332** by laser etching or mechanical drilling. The method can also include forming the conducting material layer by plating or depositing on surface of enclosure **332** and in the openings, and

forming patterned conducting layer **53** by etching the conducting material layer with a mask.
(34) Conducting layer **53** may include metal layer **531** that extends on the surface of enclosure **332**, where metal layer **531** serves as a thicker layer of conducting layer **53**. Conducting layer **53** may also include metal layer **532** disposed on metal layer **531** and in the openings of enclosure **332**, and metal layer **533** disposed on metal layer **532**. Metal layer **532** may serve as a metal seed layer for forming metal layer **533** by plating. In addition, conducting layer **53** may also include solder layer **534** disposed on metal layer **553**. Enclosure **333** may cover die **630**, in order to protect die **630** from possibly being damaged or polluted.
(35) Thus in certain embodiments, the stacked package structure may be adapted to a chip package with a relatively high space density between pads, by firstly leading out the electrodes via the interlinkage and the first redistribution body when packaging the first die. In such a case, the bonding wires may not be needed and the package resistors can be reduced. By secondly leading out the electrodes on the second die via the penetrating bodies that penetrate the first enclosure and the substrate in a similar way after packaging the first die, the chip can be packaged in a stack with a reduced packaging size and fewer pins.
(36) Referring now to FIG. **4**, shown is a flow diagram of an example method of making a stacked package structure, in accordance with embodiments of the present invention. At **802**, a substrate can be provided. At **804**, a first die can be arranged on a first surface of the substrate. The first die may have an active face and a back face opposite thereto. The back face of the first die can be relatively close to the first surface of the substrate, and the active face of the first die may be provided with pads. At **806**, a first enclosure can be formed and may cover the first die.
(37) At **808**, at least one interlinkage can be formed. The interlinkage can extend to the inside of the first enclosure, and may be electrically coupled to the pads. At **810**, at least one first redistribution body can be formed, and the first redistribution body can be electrically coupled to the interlinkage. Also, the first redistribution body can be partially exposed on the surface of the surface of the stacked package structure, in order to provide outer pins for electrically coupling to external circuitry. At **812**, at least one penetrating body can be formed. The penetrating body can penetrate the first enclosure and the substrate. At **814**, at least one electrode on the second die can be electrically coupled to the first terminal of the penetrating body. Also, the second terminal of the penetrating body may be partially exposed on the surface of the stacked package structure, in order to provide outer pins for electrically coupling to external circuitry.
(38) In particular embodiments, the stacked package structure can be adapted to a chip package with a relatively high space density between pads, by firstly leading out the electrodes via the interlinkages and the first redistribution bodies when packaging the first die. In such a case, the bonding wires may not be needed and the package resistors can be reduced. Secondly, by leading out the electrodes on the second die via the penetrating bodies that penetrate the first enclosure and the substrate after packaging the first die, the chip can be packaged in a stacked configuration with reduced packaging size and fewer pins.
(39) The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, to thereby enable others skilled in the art to best utilize the invention and various embodiments with modifications as are suited to particular use(s) contemplated. It is intended that the scope of the invention be defined by the claims appended hereto and their equivalents.

Claims

1. A stacked package structure for a chip, comprising: a) a substrate having a first surface and a second surface opposite thereto; b) a first die and a second die arranged on opposite two sides of said substrate; c) said first die having an active face and a back face opposite thereto, wherein said first die is arranged above said first surface of said substrate, said back face of said first die is

closer than said active face of said first die to said first surface of said substrate, and said active face of said first die comprises pads; d) a first enclosure that encapsulates side surfaces and said active face of said first die, wherein said substrate is in contact with said first enclosure, and said substrate excludes solder bumps and solder balls; e) at least one interlinkage that extends to said first enclosure to electrically couple to said pads; f) at least one first redistribution body electrically coupled to said interlinkage, and being partially exposed on a surface of said stacked package structure to provide outer pins for electrically coupling to external circuitry; g) at least one penetrating body that penetrates said first enclosure and said substrate; h) said second die having at least one electrode electrically coupled to a first terminal of said penetrating body, wherein said second die is arranged below said second surface of said substrate, said second die comprises an active face and a back face opposite thereto, said active face of said second die is closer than said back face of said second die to said second surface of said substrate, said active face of said second die comprises at least one electrode, and side surfaces and said back face of said second die are encapsulated by a same material; and i) a second terminal of said penetrating body that is at least partially exposed on said surface of said stacked package structure to provide outer pins for electrically coupling to said external circuitry.
2. The stacked package structure of claim 1, wherein said interlinkage comprises a first portion that extends on a surface of said first enclosure, and a second portion in said first enclosure and that extends to a corresponding pad.
3. The stacked package structure of claim 2, wherein said first redistribution body comprises: a) a first portion that extends from said surface of said first enclosure to said interlinkage and electrically couples to said interlinkage; b) a second portion extending from a surface of said first enclosure to a second surface of said substrate; and c) a third portion extending on said second surface of said substrate, wherein said first portion of said first redistribution body extends to said first portion of said interlinkage and is electrically coupled to said interlinkage, said third portion of said first redistribution is partially exposed on said surface of said stacked package structure, to provide outer pins for electrically coupling to said external circuitry, said penetrating body comprises a first terminal that extends on said surface of said first enclosure, a second terminal that extends on said second surface of said substrate, and a mid-section that extends from said surface of said first enclosure to said second surface of said substrate.
4. The stacked package structure of claim 3, wherein: a) said active face of said second die basis toward said first enclosure, and at least one electrode of said second die is electrically coupled to said first terminal of said penetrating body through conductive bumps; and b) at least one electrode on said second die is electrically coupled to said first portion of said interlinkage through said conductive bumps.
5. The stacked package structure of claim 3, wherein: a) said second die comprises a back face and an active face opposite thereto, said back face of said second die being above and separated from said first portion of said interlinkage by a conductive bump, and at least one electrode on said active face of said second die being electrically coupled to said first terminal of said penetrating body via conductive wires; and b) said at least one electrode on said active face is electrically coupled to said first portion of said interlinkage.
6. The stacked package structure of claim 3, wherein said interlinkage, said first terminal and said mid-section of said penetrating body, said first and said second portions of said first redistribution body are formed by a patterned first conducting layer comprising a first portion that extends above said first enclosure, a second portion that extends from said surface of said first enclosure to said pad, and a third portion that extends from said surface of said first enclosure to said second surface of said substrate.
7. The stacked package structure of claim 3, wherein said outer pins are formed by a patterned third conducting layer comprising a third metal and a solder layer located on said third metal layer.
8. The stacked package structure of claim 3, further comprising a second enclosure that covers said

second die.

9. The stacked package structure of claim 2, further comprising: a) a second enclosure that covers said interlinkage; b) said penetrating body having a first terminal that extends on said second surface of said substrate, a second terminal having a first portion that extends on said surface of said second enclosure and a second portion that extends to the inside of said second enclosure, and a mid-section that extends from said second terminal of said penetrating body to said second surface of said substrate; c) said first redistribution body comprises a first portion that extends on said surface of said second enclosure, and a second portion that extends to the inside of said second enclosure and is electrically coupled to said interlinkage; and d) said first portion of said first redistribution is partially exposed on said surface of said stacked package structure to provide outer pins for electrically coupling to said external circuitry.

10. The stacked package structure of claim 9, wherein said active face of said second die faces toward said second surface of said substrate, and at least one electrode on said second die is electrically coupled to said first terminal of said penetrating body via conductive bumps.

11. The stacked package structure of claim 9, wherein said second die comprises a back face and an active face opposite thereto, said back face of said second die being relatively to said first terminal of said interlinkage, and at least one electrode on said active face of said second die being electrically coupled to said first terminal of said penetrating body via conductive wires.

12. The stacked package structure of claim 9, wherein at least one second terminal of said penetrating body extends to said first portion of said first redistribution body to electrically couple to said first redistribution body.

13. The stacked package structure of claim 9, wherein: a) said mid-section of said penetrating body comprises a first portion that extends on said surface of said first enclosure and a second portion that extends to said second surface of said substrate, said mid-section of said penetrating body and said interlinkage being formed by said patterned first conducting layer, said first conducting layer comprises a first portion that extends on said first enclosure, a second portion that extends from said surface of said first enclosure to said pad, and a third portion that extends from said surface of said first enclosure to said second surface of said substrate; and b) said first conducting layer comprises a first metal layer and a second metal layer disposed on said first metal layer.

14. The stacked package structure of claim 9, wherein said second terminal of said penetrating body and said first redistribution body are formed by said patterned second conducting layer, said second conducting layer comprising a first portion that extends on said surface of said second enclosure, and a second portion that extends to the inside of said second enclosure.

15. The stacked package structure of claim 14, wherein: a) said first conducting layer comprises a third metal layer and a fourth metal layer disposed on said third metal layer, and a solder layer disposed on said surface of said fourth metal layer; and b) said first conducting layer further comprises a thicker layer under said third metal layer, wherein said thicker layer extends on said surface of said second enclosure.

16. The stacked package structure of claim 9, wherein said first terminal of said penetrating body is formed by said patterned third conducting layer, and further comprising a third enclosure for covering said second die.

17. A method of making a stacked package structure, the method comprising: a) providing a substrate; b) arranging a first die on a first surface of said substrate, wherein said first die comprises an active face and a back face opposite thereto, said back face of said first die is closer than said active face of said first die to said first surface of said substrate, and said active face of said first die comprises pads; c) forming a first enclosure to encapsulate side surfaces and said active face of said first die, wherein said substrate is in contact with said first enclosure, and said substrate excludes solder bumps and solder balls; d) forming at least one interlinkage that extends to the inside of said first enclosure and is electrically coupled to said pads; e) forming at least one first redistribution body, and electrically coupling said first redistribution body to said interlinkage, wherein said first

redistribution body is partially exposed on said surface of said stacked package structure to provide outer pins for electrically coupling to external circuitry; f) forming at least one penetrating body that penetrates said first enclosure and said substrate; and g) electrically coupling at least one electrode on a second die to said first terminal of said penetrating body, wherein said second terminal of said penetrating body is partially exposed on said surface of said stacked package structure to provide outer pins for electrically coupling to said external circuitry, wherein said first die and said second die are arranged on opposite two sides of said substrate, h) wherein said second die is arranged below said second surface of said substrate, said second die comprises an active face and a back face opposite thereto, said active face of said second die is closer than said back face of said second die to said second surface of said substrate, said active face of said second die comprises at least one electrode, and side surfaces and said back face of said second die are encapsulated by a same material.